

Bachelor in Computer Applications

Course Title: **Database Management System**

Code No: Semester: **3**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Note: Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.

Group B

1. Define serial, non-serial and serializable schedules with example. How can you test serializability of a schedule? Explain with an example
2. What is Transaction? State and explain the states of transaction with transition diagram.
3. From the the relations given below, answer the following questions.EMPLOYEE (Ssn, Fname, Bdate, Address, Sex, Salary, Super_ssn, Dno)DEPARTMENT (Dname, Dnumber, Mgr_ssn, Mgr_start_date)PROJECT (Pname, Pnumber, Plocation, Dnum)DEPENDENT (Essn, Dependent_name, Sex, Bdate, Relationship)DEPT_LOCATIONS (Dnumber, Dlocation)a) Retrieve the name and address of all employees who work for the 'Computer' department in SQLb) For each Department, retrieve the department number, the number of employees in the department, and their average salary using SQL.c) Retrieve the employees name and their Dname and Pname ordered by the employee's Dname.d) Retrieve the name and salary of all employees who work in department number 5 using Relational Algebra.e) Retrieve the name of the manager of each department using Relational Algebra.
4. What are the Advantages of using DBMS? Explain in brief.
5. What is Attributes? Explain different types of Attributes with example.
6. Define relation, instance and schema with example.
7. What is Normalization? Explain the 2NF with Examples.
8. Wha is Shared/Exclusive (Read/Write) Locks? How it is different from binary Locks.
9. What is Deadlock in DBMS? Explain with example.
10. What is shadow paging? How it is used for database recovery?
11. What is data independence? How three schema architecture ensures logical and physical data independence?
12. What is NoSQL? Explain the characteristics of NoSQL.